

HW 3.

1. (a) c-class ; k features iid $x_j | u_i \sim \text{Geom}(p_i d_j)$

Bayes's optimal classifier's decision :

$$p(u_i | x) \propto p(x | u_i) p(u_i)$$

We need to choose $\max p(u_i | x)$

$$\Rightarrow \hat{u}(x) = \arg \max_{u_i} p(x | u_i) p(u_i)$$

$$\text{We have : } p_{x_j | u_i}(x_j | u_i) = \frac{1}{T(p_i)} \lambda_j^{p_i} x_j^{p_i-1} e^{-\lambda_j x_j} \quad p_i, \lambda_j > 0$$

$$\Rightarrow p(x | u_i) = \prod_{j=1}^k p(x_j | u_i)$$

$$\begin{aligned} \Rightarrow p(x | u_i) p(u_i) &= \prod_{j=1}^k p(x_j | u_i) p(u_i) \\ &= \left(\frac{1}{T(p_i)}\right)^k \left(\prod_{j=1}^k \lambda_j^{p_i}\right) \left(\prod_{j=1}^k x_j^{p_i-1}\right) \exp\left(-\sum_{j=1}^k \lambda_j x_j\right) p(u_i) \end{aligned}$$

$$\Rightarrow \hat{u}(x) = \arg \max_{u_i} \left[p(u_i) \left(\frac{1}{T(p_i)}\right)^k \left(\prod_{j=1}^k \lambda_j^{p_i}\right) \left(\prod_{j=1}^k x_j^{p_i-1}\right) \right]$$

Take the logarithm :

$$g_i(x) = \ln p(u_i) - k \ln T(p_i) + p_i \sum_{j=1}^k \ln \lambda_j + (p_i - 1) \sum_{j=1}^k \ln x_j$$

Therefore we get that decision :

$$\hat{u}(x) = \arg \max_{u_i} g_i(x)$$

(b) let $g_0(x) = \ln p(u_0) - k \ln T(p_0) + p_0 \sum_{j=1}^k \ln \lambda_j + (p_0 - 1) \sum_{j=1}^k \ln x_j$

$$g_1(x) = \ln p(u_1) - k \ln T(p_1) + p_1 \sum_{j=1}^k \ln \lambda_j + (p_1 - 1) \sum_{j=1}^k \ln x_j$$

the decision boundary is $g_1(x) = g_0(x)$

$$g_0(x) - g_1(x) = (p_0 - p_1) \sum_{j=1}^k \ln x_j + T = 0$$

$T =$ a constant. It is independent of x .

$\Rightarrow$ therefore $(p_0 - p_1) \sum_{j=1}^k \ln x_j + T$ is linear to $\ln(x_j)$

(c) $p_1 = 4 \quad p_2 = 2 \quad c = 2 \quad k = 4 \quad \lambda_1 = \lambda_3 = 1 \quad \lambda_2 = \lambda_4 = 2$
 $X = (0.1, 0.2, 0.3, x)$

from (a) $\Rightarrow g_i(x) = \ln p(u_i) - k \ln T(p_i) + p_i \sum_{j=1}^k \ln \lambda_j + (p_i - 1) \sum_{j=1}^k \ln x_j$

$$\sum_{j=1}^4 \ln(\lambda_j) = \ln 1 + \ln 2 + \ln 1 + \ln 2 = 2 \ln 2$$

$$\sum_{j=1}^4 \ln(x_j) = \ln(0.1) + \ln(0.2) + \ln(0.3) + \ln(4) \approx -3.73$$

$$\Rightarrow g_i(x) = \ln p(u_i) - 4 \ln T(p_i) + 2 \ln 2 \cdot p_i + (p_i - 1) (-3.73)$$

$$g_1(x) = \ln 4 - 4 \ln 6 + 2 \ln 2 \times 4 + 3 \times -3.73 \approx -11.428$$

$$g_2(x) = \ln 2 - 4 \ln 1 + 2 \ln 2 \times 2 + 1 \times (-3.73) \approx -0.265$$

therefore $X \Rightarrow W_2$, because $g_2(x) > g_1(x)$

d, $p_1 = 3.2$ $p_2 = 8$ $c=2$ $k=1$ $\lambda_1=1$

$$\Rightarrow X|u_1 \sim \text{Gamma}(3.2, 1) \quad X|u_2 \sim \text{Gamma}(8, 1)$$

$$g_i(x) = \ln p(x|u_i) = -\ln \Gamma(p_i) + (p_i-1) \ln x - x$$

$$\text{Let } g_1(x) = g_2(x) \Rightarrow -\ln \Gamma(3.2) + 2.2 \ln x = -\ln \Gamma(8) + 7 \ln x$$

$$\Rightarrow x^* = \left(\frac{\Gamma(8)}{\Gamma(3.2)} \right)^{1/4.8} \approx 4.8$$

$$\text{Therefore } x < x^* \Rightarrow u_1, \quad x > x^* \Rightarrow u_2$$

$$P(\text{Type I}) = P(X > x^* | u_1) = 1 - F_{\Gamma(3.2, 1)}(x^*) \approx 0.157$$

$$P(\text{Type II}) = P(X < x^* | u_2) = F_{\Gamma(8, 1)}(x^*) \approx 0.124$$

(e) $p_1 = p_2 = 4$; $c=2$; $k=2$, $\lambda_1=8$, $\lambda_2=0.3$

$$p(u_1) = 1/4 \quad p(u_2) = 3/4$$

$$\Rightarrow p(x|u_1) = p(x|u_2)$$

$$\therefore p(u_1|x) \propto p(x|u_1)p(u_1)$$

$$\therefore p(u_2) > p(u_1)$$

$$\Rightarrow f(x, x_2) = \frac{1/4 p(x|u_1) p(u_1)}{1/4 p(x|u_2) p(u_2)} = 1/4 \frac{(1/4)}{(3/4)}$$

$$\text{Therefore } f(x, x_2) = 1/12 < 0$$

$$2. \quad x_i | w_j \sim \text{lap}(m_{ij}, \lambda_i) \quad p(x_i | w_j) = \frac{1}{\lambda_i} e^{-\lambda_i |x_i - m_{ij}|}$$

$$\text{from Bayes} \quad \hat{w}(x) = \arg \max_j p(x | w_j)$$

$\therefore$ iid

$$\therefore p(x | w_j) = \left(\prod_{i=1}^k \frac{1}{\lambda_i} \right) \exp\left(-\sum_{i=1}^k \lambda_i |x_i - m_{ij}|\right)$$

$$\hat{w}(x) = \arg \max_j \exp\left(-\sum_{i=1}^k \lambda_i |x_i - m_{ij}|\right)$$

$$\hat{w}(x) = \arg \min_j \underbrace{\sum_{i=1}^k \lambda_i |x_i - m_{ij}|}_{\text{Weighted L1 distance}}$$

Weighted L1 distance.

$$\text{When } \lambda_1 = \lambda_2 = \dots = \lambda_k = 1 \quad \sum_{i=1}^k \lambda_i |x_i - m_{ij}| = \sum_{i=1}^k |x_i - m_{ij}|$$

$$\Rightarrow \hat{w}(x) = \arg \min_j \sum_{i=1}^k |x_i - m_{ij}|$$

$$3. \quad (a) \quad R(\alpha_j | x) = \sum_{j=1}^4 \lambda(\alpha_j | w_j) p(w_j | x)$$

$$p(w_j | x) = \frac{p(x | w_j) p(w_j)}{\sum_{k=1}^4 p(x | w_k) p(w_k)}$$

$$p(w_1) = 1/10 \quad p(w_2) = 1/5 \quad p(w_3) = 1/2 \quad p(w_4) = 1/5$$

When $x=1$

$$p(w_1 | 1) = \frac{10}{89}, \quad p(w_2 | 1) = \frac{30}{89}, \quad p(w_3 | 1) = \frac{25}{89}, \quad p(w_4 | 1) = \frac{24}{89}$$

$$\Rightarrow R(\alpha_1 | 1) = 0 \times \frac{10}{89} + 2 \times \frac{30}{89} + 3 \times \frac{25}{89} + 4 \times \frac{24}{89} = \frac{231}{89}$$

$$R(\alpha_2 | 1) = \frac{237}{89} \quad R(\alpha_3 | 1) = \frac{138}{89} \quad R(\alpha_4 | 1) = \frac{165}{89}$$

When $X=2$

$$P(W_1|2) = \frac{10}{99} \quad P(W_2|2) = \frac{15}{99} \quad P(W_3|2) = \frac{10}{99} \quad P(W_4|2) = \frac{24}{99}$$

$$R(\alpha_1|2) = \frac{276}{99} \quad R(\alpha_2|2) = \frac{252}{99} \quad R(\alpha_3|2) = \frac{108}{99} \quad R(\alpha_4|2) = \frac{165}{99}$$

When $X=3$

$$P(W_1|3) = \frac{10}{112} \quad P(W_2|3) = \frac{15}{112} \quad P(W_3|3) = \frac{25}{112} \quad P(W_4|3) = \frac{12}{112}$$

$$R(\alpha_1|3) = \frac{302}{112} \quad R(\alpha_2|3) = \frac{181}{112} \quad R(\alpha_3|3) = \frac{86}{112} \quad R(\alpha_4|3) = \frac{170}{112}$$

Therefore, from optimal Bayesian decision, choose α_3

$$\begin{aligned} (b) \quad R &= \sum_{i=1}^3 R(\alpha_3|\pi_i) P(\pi_i) \\ &= R(\alpha_3|1)P(1) + R(\alpha_3|2)P(2) + R(\alpha_3|3)P(3) \end{aligned}$$

$$P(1) = \frac{1}{3} \times \frac{1}{10} + \frac{1}{2} \times \frac{1}{5} + \frac{1}{6} \times \frac{1}{2} + \frac{2}{5} \times \frac{1}{5} = \frac{89}{200}$$

$$P(2) = \frac{1}{3} \times \frac{1}{10} + \frac{1}{4} \times \frac{1}{5} + \frac{1}{3} \times \frac{1}{2} + \frac{2}{5} \times \frac{1}{5} = \frac{99}{200}$$

$$P(3) = \frac{1}{5} \times \frac{1}{10} + \frac{1}{4} \times \frac{1}{5} + \frac{1}{2} \times \frac{1}{2} + \frac{1}{5} \times \frac{1}{5} = \frac{112}{200}$$

$$R(\alpha_3|1) = \frac{138}{89} \quad R(\alpha_3|2) = \frac{12}{11} \quad R(\alpha_3|3) = \frac{3}{4}$$

$$R = \frac{138}{89} \times \frac{89}{200} + \frac{12}{11} \times \frac{99}{200} + \frac{3}{4} \times \frac{112}{200} = \frac{11}{10}$$

Therefore: $R = \frac{11}{10}$

4.

(a) let $w_1 = \text{Yes}$ $w_0 = \text{No}$.

$$P(w_1) = \frac{3}{10} \quad P(w_0) = \frac{7}{10}$$

(b) only one feature is not relevant $\Rightarrow$ it's should be Tax ID

$$P(X|w_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(X-\mu_i)^2}{2\sigma_i^2}\right)$$

$$\text{MLE: } \hat{\mu}_i = \frac{1}{n} \sum x \quad \hat{\sigma}_i^2 = \frac{1}{n} \sum (x - \hat{\mu}_i)^2$$

$$\text{when } w_0 = \text{No} \quad \mu_0 = (122 + 77 + 106 + 210 + 72 + 111 + 60) / 7 \approx 101.14$$

$$\hat{\sigma}_0^2 = \frac{1}{7} \times \frac{106658}{7} = \frac{106658}{49} \approx 2176.69$$

$$\text{when } w_1 = \text{Yes} \quad \mu_1 = \frac{263}{3} \approx 87.67 \quad \hat{\sigma}_1^2 = \frac{18}{9} \approx 2$$

$$\Rightarrow P(X|w_0) = N(101.14, 2176.69) \quad P(X|w_1) = N(87.67, 2)$$

$$(c) P_{X|w_i}(x_j|w_i) = P(X=x_j|w_i) = \frac{n_{ji}}{n_i}$$

$$P(\text{Refund} = \text{Yes} | w_1) = 0 \quad P(\text{Refund} = \text{No} | w_1) = 1$$

$$P(\text{Refund} = \text{Yes} | w_0) = 3/7 \quad P(\text{Refund} = \text{No} | w_0) = 4/7$$

$$P(\text{Single} | w_1) = 2/3 \quad P(\text{Married} | w_1) = 0 \quad P(\text{Divorced} | w_1) = 1/3$$

$$P(\text{Single} | w_0) = 2/7 \quad P(\text{Married} | w_0) = 4/7 \quad P(\text{Divorced} | w_0) = 1/7$$

$$\therefore 0 \leq P(X=x_j|w_i) \leq 1 \Rightarrow 0 \leq \frac{n_{ji}}{n_i} \leq 1$$

$$\sum_j P(X=x_j|w_i) = 1 \quad \sum_j n_{ji} = n_i$$

Therefore: $\sum_j \frac{n_{ji}}{n_i} = \frac{1}{n_i} \sum_j n_{ji} = \frac{n_i}{n_i} = 1$, it's a valid pmf.

(d) from (c) $P(X=x_j|w_i) = \frac{n_{ji}}{n_i}$ will have zero probability

So, we can use Laplace correction: $P(X=x_j|w_i) = \frac{n_{ji}+1}{n_i+L}$

add a small nonzero probability to every possible feature value.

This can avoid zero probabilities.

$$\sum_{j=1}^L \frac{n_{ji}+1}{n_i+L} = \frac{n_i+L}{n_i+L} = 1$$

Therefore, this will be a valid pmf.

(e) $X = (\text{Refund}, \text{Marital Status}, \text{Income})$

$$\hat{w}(x) = \arg \max_{w_i} p(w_i) p(x|w_i)$$

$$p(w_i|x) \propto p(w_i) \cdot p(\text{Income}|w_i) \cdot p(\text{Marital}|w_i) \cdot p(\text{Refund}|w_i)$$

$$p_{w_1} = 0.3 \quad p_{w_2} = 0.7$$

Refund Laplace $L=2$

$$p(\text{Refund} = \text{Yes} | w_1) = \frac{1}{5} \quad p(\text{Refund} = \text{No} | w_1) = \frac{4}{5}$$

$$p(\text{Refund} = \text{Yes} | w_2) = \frac{4}{9} \quad p(\text{Refund} = \text{No} | w_2) = \frac{5}{9}$$

Marital Status $L=3$

$$p(\text{Single} | w_1) = \frac{1}{2} \quad p(\text{Married} | w_1) = \frac{1}{6} \quad p(\text{Divorced} | w_1) = \frac{1}{3}$$

$$p(\text{Single} | w_2) = \frac{3}{10} \quad p(\text{Married} | w_2) = \frac{1}{2} \quad p(\text{Divorced} | w_2) = \frac{1}{5}$$

In case:

$$p(x|w_1) = N(87.67, 4.4)$$

$$p(x|w_0) = N(109.14, 2176.68)$$

$\Rightarrow$ therefore classify as Evade = Yes if

$$\frac{3}{10} p(\text{Refund}|w_1) p(\text{Marital}|w_1) p(\text{Income}|w_1) >$$

$$\frac{7}{10} p(\text{Refund}|w_0) p(\text{Marital}|w_0) p(\text{Income}|w_0)$$

otherwise classify Evade = No.